Ziao Yang

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EDUCATION

Brandeis University — Waltham, MA
Ph.D. Candidate in Computer Science — 09/2023 - 05/2027 (Expected)
Advisor: Professor Hongfu Liu

Sun Yat-Sen University — Guangzhou, China
B.Eng. in Computer Science and Technology — 09/2018 - 06/2022

RESEARCH INTERESTS

Large Language Models: Data Curation for Pre-Training and Post-Training, AI Agents, Model Evaluation, and Efficient Inference

Data-Centric Learning: Data Valuation, Influence Functions, and Data-Efficient Training

Spatiotemporal Forecasting: Transformer-based Video Prediction

SKILLS

Python, PyTorch, JAX, Hugging Face, Tunix, DeepSpeed, Multi-GPU, Multi-Host TPU, vLLM, SGLang, SFT, PPO, GRPO, DPO, Distillation, Reward Modeling, Reward Design, Data Curation, Synthetic Data, AI Agents, RL Environment Design, LLM Evaluation, Agent Evaluation, Verifiers & Graders, Eval Harnesses

RESEARCH & INDUSTRY EXPERIENCE

Adobe Research — Seattle, WA
Research Scientist Intern, Mentor: Dr. Vlad Morariu — 05/2026 - 08/2026
Worked on accelerating HTML generation in Adobe's Project Avalon agentic pipeline by replacing Claude Sonnet calls in selected agents with locally served LLMs.

- Developed task decomposition and parallel generation methods, evaluating the trade-offs between inference latency and output quality.
- Explored knowledge distillation; built an end-to-end pipeline to generate data with a 480B teacher and fine-tune, serve, and evaluate 8B student LLMs.

Google Tunix Research Collaboration — Remote
Academic Research Collaborator — 09/2025 - 05/2026
Researched data-centric LLM post-training using Google Tunix on TPUs.

- Developed self-influence and outlier-based methods to filter low-signal samples in GRPO and DPO.
- Implemented DeepScaleR math-reasoning post-training and AIME 2024 evaluation on multi-host TPUs.

Adobe — San Jose, CA
Machine Learning Engineer Intern, Mentor: Dr. Lei Zhang — 05/2025 - 08/2025
Developed controllable headline and email generation for Adobe GenStudio for Performance Marketing, supporting enterprise marketing workflows.

- Improved offline quality through multi-persona generation and evaluation.
- Trained a reward model and applied PPO for brand-aligned generation; first-author paper accepted to EMNLP 2026 (main conference).

International Digital Economy Academy (IDEA) — Shenzhen, China
Research Scientist Intern, Mentor: Dr. Jiaxing Zhang — 03/2022 - 07/2022
Contributed to the Fengshenbang language-model project (4.1K GitHub stars).

- Pre-trained BERT (1.3B) and GPT-2 (3.5B) models from scratch.
- Built datasets of 27 million labeled examples for multi-task classification.
- Performed supervised continued pre-training across classification tasks using MAML and Reptile.

RESEARCH & INDUSTRY EXPERIENCE

Y Combinator China — Beijing, China
Intern, Mentor: Dr. Qi Lu — 03/2023 - 07/2023

Conducted AI industry research and startup scouting, translating technical and market trends into strategic insights for leadership.

- Prepared presentations and financing materials for Dr. Qi Lu, including "My Foundation Model Worldview."
- Scouted emerging AI startups and synthesized insights from discussions with scientists at leading AI labs.

Sun Yat-Sen University — Guangzhou, China
Research Assistant, Mentor: Dr. Qifeng Lin — 2020 - 2022

Researched Transformer-based video and spatiotemporal forecasting.

- Developed TCTN, combining Transformer attention with 3D temporal convolutions to model short- and long-range dependencies for future-frame prediction on MovingMNIST and KTH.
- Extended TCTN to PTCT for autoregressive radar-echo forecasting and precipitation nowcasting on Radar Echo Guangzhou and HKO-7.

SELECTED PUBLICATIONS

Ziao Yang, Han Yue, Jian Chen, and Hongfu Liu. "Revisit, Extend, and Enhance Hessian-Free Influence Functions." *Transactions on Machine Learning Research* (**TMLR**), 2026. [PDF] [Code]

Ziao Yang, Zizhang Chen, Lei Zhang, and Hongfu Liu. "Recontextualizing Famous Quotes for Brand Slogan Generation." *Conference on Empirical Methods in Natural Language Processing* (**EMNLP**), 2026. [PDF] [Code]

Xuesong Jia*, **Ziao Yang***, Zhanhe Huang, and Hongfu Liu. "Not All Rollouts Are Worth Learning: On Trajectory Valuation for Post-Training Reinforcement Learning." Submitted to *International Conference on Learning Representations* (**ICLR**), 2027. *arXiv preprint* arXiv:2609.35072. *(* Co-first authors.)* [PDF] [Code]

Ziao Yang, Longbo Huang, and Hongfu Liu. "Layer-Aware Influence for Online Data Valuation Estimation." Submitted to *IEEE Transactions on Pattern Analysis and Machine Intelligence* (**TPAMI**), 2026. [PDF] [Code]

Ziao Yang, Han Yue, Jian Chen, and Hongfu Liu. "On the Inflation of KNN-Shapley Value." Submitted to *Data-centric Machine Learning Research* (**DMLR**), 2026. [PDF] [Code]

Yinan Shen, **Ziao Yang**, and Hongfu Liu. "Validation-Induced Shapley Shifts: How Validation Structure Distorts Data Valuation." *arXiv preprint* arXiv:2607.03675, 2026. [PDF]

Ziao Yang, Xiangrui Yang, and Qifeng Lin. "TCTN: A 3D-Temporal Convolutional Transformer Network for Spatiotemporal Predictive Learning." *arXiv preprint* arXiv:2112.01085v1, 2021. [PDF] [Code]

Ziao Yang, Xiangrui Yang, and Qifeng Lin. "PTCT: Patches with 3D-Temporal Convolutional Transformer Network for Precipitation Nowcasting." *arXiv preprint* arXiv:2112.01085v2, 2022. [PDF]

OPEN SOURCE

TCTN – First author; official implementation	97 GitHub stars
Fengshenbang-LM – Contributor	4.1K GitHub stars
Agents' Last Exam – Contributor	1.1K GitHub stars
ClawBench – Contributor	1.1K GitHub stars

PROFESSIONAL ACTIVITIES

Reviewer

Neural Information Processing Systems (**NeurIPS**)	2024 - Present
International Conference on Learning Representations (**ICLR**)	2024 - Present
International Conference on Machine Learning (**ICML**)	2024 - Present

HONORS & AWARDS

Tianchi Big Data Competition - AI Helps Weather Forecast (Top 1%) — 2022
Alibaba Group

National Level of College Student Innovation Competition (Top 1) — 2022
Sun Yat-Sen University